

BRAR Annotation Protocol v1.0

Bone Resorption Age Ratio – Standardized Grading Protocol for Panoramic Radiographs

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1. Image Acquisition Protocol

All panoramic radiographs were acquired using the **OC200D Panoramic Radiography System** (DEXIS, Envista, USA) under standardized clinical conditions.

1.1 Patient Positioning

- **Midsagittal plane:** Perpendicular to the floor.
- **Frankfort horizontal plane:** Parallel to the floor.
- **Anterior teeth:** Centered within the focal trough using the manufacturer-provided bite block and light guides.
- **Tongue position:** Patients instructed to press the tongue firmly against the palate to minimize palatoglossal air space.

1.2 Retake Policy

A single retake was permitted if any of the following positioning errors were observed:

- Incisor foreshortening or elongation (indicating off-focal placement)
- Severe motion blur
- Excessive palatoglossal air space
- Premolar overlap exceeding one cusp width

Images failing to meet minimal diagnostic criteria after one retake were **excluded** from BRAR analysis.

2. Anatomical Landmark Definitions

Accurate landmark identification is critical for BRAR calculation. All annotators followed these radiographic definitions:

2.1 Cemento-Enamel Junction (CEJ)

- Defined as the **most apical point of the enamel–cementum interface** on the mesial or distal surface of the tooth.
- **Avoid restorative margins** (e.g., fillings, crowns); if CEJ is obscured by restoration, use the nearest identifiable enamel edge.
- If CEJ cannot be localized within **±1.5 mm confidence**, the site is flagged for consensus review.

2.2 Alveolar Crest

- Defined as the **most coronal interproximal bone peak** between two adjacent teeth.
- **Exclude lamina dura projections** or sharp bony spikes not representing true crestal bone.
- In edentulous regions or isolated teeth, crest is not defined.

 **Exclusion Rule:** Any tooth where either CEJ or alveolar crest cannot be confidently identified within ± 1.5 mm is **excluded from quantitative BL measurement** but may be retained for categorical annotation (e.g., missing, implant).

3. Bone Resorption Quantification

3.1 Bone Loss (BL)

- Measured in **millimeters (mm)** as the vertical distance from the CEJ to the alveolar crest on the **tooth with the greatest bone loss** per patient.
- Only interproximal sites are used (mesial/distal).

3.2 Root Length (RL)

- Measured in **millimeters (mm)** as the distance from the CEJ to the **root apex** of the same tooth used for BL measurement.

3.3 BRAR Formula

The **Bone Resorption Age Ratio (BRAR)** is a unitless, age-normalized severity index:

$$BRAR = \frac{Age \cdot BL}{RL}$$

Where:

- **BL:** Bone loss (mm) at the most affected tooth
- **RL:** Root length (mm) of the same tooth
- **Age:** Patient’s chronological age (years) at time of imaging

 **Note:** Per the 2018 AAP Periodontal Classification, severity is anchored on the **most affected site**, not whole-mouth average. Thus, patient-level BRAR uses the **maximum BL/RL ratio** across all teeth.

4. Severity Grading System

Based on BRAR, periodontal bone resorption is categorized into three clinical grades:

1	< 0.25	Mild bone resorption
2	0.25 – 1.0	Moderate bone resorption
3	> 1.0	Severe bone resorption

This grading enables cross-patient comparison of bone remodeling burden independent of age.

5. Annotation Workflow & Quality Control

5.1 Annotator Team

- **3 junior periodontists:** Independently performed initial annotations after calibration.
- **2 senior periodontists:** Conducted final adjudication and resolved discrepancies.

5.2 Training & Calibration

- All annotators underwent training using **pilot cases** and **visual exemplars** of CEJ/crest identification and BRAR calculation.
- Inter-rater agreement was assessed on a **held-out subset (n = 50)** prior to full-scale annotation.

5.3 Consensus Process

1. Each case independently graded by 3 annotators.
2. Disagreements in severity grade resolved by **majority vote**.
3. Unresolved cases escalated to **senior periodontist adjudication**.
4. Final dataset reflects **consensus labels** reviewed by senior specialists.

5.4 Technical Validation

Inter-rater reliability (Cohen's κ / ICC) is reported in the associated *Scientific Data* publication under "Technical Validation".

6. Data Exclusion Criteria Summary

A tooth or patient is excluded from BRAR calculation if:

- Image fails diagnostic quality after one retake
- CEJ or alveolar crest cannot be localized within ± 1.5 mm
- Root apex not visible (prevents RL measurement)
- Patient age missing or < 18 years

 All included data are **fully anonymized**; no protected health information is retained.

End of Protocol

This document accompanies the BRAR dataset (DOI: 10.6084/m9.figshare.30155974)